Chapter 1 Knowledge: Introduction to Electromagnetic Compatibility

Outline

- Preview / Basic Description of EMC
- Aspects of EMC
- History of EMC
- Examples
- Electrical Dimensions and Waves
- Decibels and Common EMC Units

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1. Preview — Basic Description of EMC

Categories of Receiver Interference *(chap1.pdf, p.3)*

- **Intentional Receiver**: A receiver designed to receive signals (e.g., AM radio) may receive unintended signals (e.g., noise from lightning, or transmissions it is not tuned to).
- **Unintentional Receiver**: A device not designed to receive signals (e.g., a digital computer) may nonetheless pick up strong FM/TV transmissions, or conversely radiate emissions that interfere with nearby devices (e.g., TV screens).

Definition of EMC *(chap1.pdf, p.4)*

> An electronic system is said to be **electromagnetically compatible** with its environment if it does not produce or be susceptible to interference. (It may couple a signal into the system, but does not cause interference.)

Criteria of EMC *(chap1.pdf, p.4)*

1. It does **not cause interference** with other systems.
2. It is **not susceptible** to emissions from other systems.
3. It does **not cause interference with itself**.

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2. Aspects of EMC

Basic Decomposition of EMC Coupling *(chap1.pdf, p.5)*

EMC is concerned with the **generation, transmission, and reception** of electromagnetic energy.

The three-element model:

Source (emitter) → Transfer (coupling) path → Receptor (receiver)

(chap1.pdf, p.5)

- Interference occurs if the **received energy** causes the receptor to behave in an **undesired manner**.
- Unintentional transmission or reception is **not necessarily detrimental**; only undesired behavior of the receptor constitutes interference.

Ways to Prevent Interference *(chap1.pdf, p.6)*

1. **Suppress the emission at its source** (most effective, least costly).
2. Make the **coupling path as inefficient** as possible for the noise.
3. Make the **receptor less susceptible** to the emission.

   Working from step 1 to step 3, success will usually be easier to achieve and with less additional design cost.

The simplest (but costliest) solution would be to enclose all electronics in a metallic enclosure with internal batteries.

Basic EMC Subproblems — Transfer of Electromagnetic Energy *(chap1.pdf, p.7)*

#	Type	Description
1	Radiated emissions	Noisy component radiates energy out
2	Radiated susceptibility	Susceptible component receives radiated energy
3	Conducted emissions	Noisy component injects noise onto power/signal lines
4	Conducted susceptibility	Susceptible component affected by conducted noise

Figure description: Four diagrams (a)–(d) show: (a) radiation from a noisy component on a PCB; (b) external radiation coupling into a susceptible component; (c) conducted noise injected back through cables; (d) conducted susceptibility from power lines.

Importance of Power Net *(chap1.pdf, p.8)*

Although a power cable is intended to carry 60 Hz commercial power, **higher-frequency signals commonly exist on the AC power cord**. These can interfere via conduction or radiation. Reducing power-net interference is a key EMC topic.

Other Aspects of EMC *(chap1.pdf, p.9)*

These are **not restricted by government** regulations but must be considered during design:

1. **Electrostatic Discharge (ESD)** — e.g., person walking on carpet touching a computer.
2. **Electromagnetic Pulse (EMP)** — e.g., from nuclear detonation affecting aircraft/facilities.
3. **Lightning** — e.g., 50,000 A strike coupling into power lines.
4. **TEMPEST** — unintentional electromagnetic emanations that can reveal processed information.

Solutions for EMC — Mathematical Model *(chap1.pdf, p.10)*

A **mathematical model** is the primary tool for understanding interference effects. A model is valid if it can **predict experimentally observed results**. Once established, it enables prediction of future events in the design process, reducing costs.

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3. History of EMC

Key Organizations *(chap1.pdf, p.11)*

Region	Organization	Year	Action
Europe	**CISPR** (International Special Committee on Radio Interference)	1933	Produced measurement equipment document for EMI emissions
USA	**FCC** (Federal Communications Commission)	1979	Published regulations requiring emissions of "digital devices" to be below limits
USA	Military	—	Sets susceptibility requirements

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4. Examples of EMI Events *(chap1.pdf, p.12–14)*

Textbook Events

- Motor spark-gap → Power cord → TV antenna → TV screen interference
- Copying machine DC transient → Power net → Hall clock reset
- Illegal FM transmitter → Microprocessor-controlled fuel system shutdown
- Citizens Band (CB) transmitter → Car braking system locked up
- Surveillance radar → Computer data loss/corruption
- Falkland Battle (1982): UK communication system interfered with antimissile detection system, destroyer lost
- EM emissions → UH-60 Black Hawk helicopter flight control malfunction
- Ship high-power radar → Inadvertent missile deployment

Daily Life Events *(chap1.pdf, p.14)*

- Mobile phone connecting to base station → Strongest EM wave → Interferes with speakers or CRT monitors
- Driving across radio station coverage areas → Favorite station disturbed by local stations

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5. Electrical Dimensions and Waves

Why Electrical Dimensions Matter *(chap1.pdf, p.15)*

- **Electrical dimensions** (in wavelengths) are more significant than physical dimensions in determining a structure's ability to radiate EM energy.
- If electrical dimensions are **small compared to** $\lambda/10$: lumped elements and circuit theory suffice.
- If electrical dimensions are **large compared to** λ: Maxwell's equations must be used.

Time Delay *(chap1.pdf, p.16)*

The time delay for a wave propagating from node a to node b along a connection lead of length $\mathscr{L}$:

$$T_D = \frac{\mathscr{L}}{v}$$

(chap1.pdf, p.16)

Figure description: Circuit showing source $i_1(t)$ connecting through a "Connection lead → Lumped element → Connection lead" to $i_2(t)$, with total length $\mathscr{L}$. The time-domain plots show $i_1(t)$ and $i_2(t)$ offset by $\mathscr{L}/v$.

Phase Difference and Electrical Length ***(chap1.pdf, p.17)***

For a sinusoidal propagating wave:

$$i(z,t) = I\cos(\omega t - \beta z)$$

(chap1.pdf, p.17)

where β is the **phase constant** and ω is the radian frequency. For a wave traveling length $\mathscr{L}$, the **phase difference** is:

$$\phi = \beta\mathscr{L} \quad \text{radians}$$

(chap1.pdf, p.17)

Since $\beta\lambda = 2\pi$ radians, the wave can be rewritten in terms of **electrical length**:

$$i(z,t) = I\cos\left(\omega t - 2\pi\frac{z}{\lambda}\right)$$

(chap1.pdf, p.17)

Figure description: Sinusoidal wave $i(0,z)$ plotted along z, showing one full wavelength λ.

Phase Velocity ***(chap1.pdf, p.18)***

Since $\omega t - \beta z =$ constant, the wave propagates in the $+z$ direction. The **phase velocity** is:

$$v = \frac{dz}{dt} = \frac{\omega}{\beta} = \lambda f \quad \text{m/s}$$

(chap1.pdf, p.18)

Therefore:

$$\lambda = \frac{v}{f} \quad \text{m}$$

(chap1.pdf, p.18)

The current with phase delay:

$$i(z,t) = I\cos\left(\omega\left(t - \frac{z}{v}\right)\right)$$

(chap1.pdf, p.19)

The **phase shift** of a wave is equivalent to a **time delay** of z/v seconds.

Phase Velocity in a Medium ***(chap1.pdf, p.20)***

$$v = \frac{1}{\sqrt{\epsilon\mu}} = \frac{v_0}{\sqrt{\epsilon_r \mu_r}}$$

(chap1.pdf, p.20)

where v_0 is the speed of light in free space:

$$v_0 = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 3 \times 10^8 \text{ m/s (approx)}$$

$$\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \text{ F/m (approx)}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m (exact)}$$

(chap1.pdf, p.20)

In a denser medium (larger ϵ_r), the velocity is smaller.

Wavelength and Electrical Dimension ***(chap1.pdf, p.21)***

Wavelength in a medium:

$$\lambda = \frac{v}{f} = \frac{v_0}{f\sqrt{\epsilon_r \mu_r}} = \frac{\lambda_0}{\sqrt{\epsilon_r \mu_r}}$$

(chap1.pdf, p.21)

Electrical dimension of a circuit:

$$k = \frac{\mathscr{L}}{\lambda}$$

(chap1.pdf, p.21)

In a denser medium: v is smaller $\Rightarrow$ λ is smaller $\Rightarrow$ k is larger.

Frequency–Wavelength Table ***(chap1.pdf, p.21)***

Frequency	Wavelength
60 Hz	3107 miles (5000 km)
3 kHz	100 km
30 kHz	10 km
300 kHz	1 km

Frequency	Wavelength
3 MHz	100 m
30 MHz	10 m
300 MHz	**1 m**
3 GHz	10 cm
30 GHz	1 cm
300 GHz	1 mm

Frequency Bands of Electronic Systems ***(chap1.pdf, p.22)***

Band	Wavelength	Uses
EHF (30–300 GHz)	1 cm–1 mm	Radar, remote sensing, radio astronomy
SHF (3–30 GHz)	10 cm–1 cm	Radar, satellite, microwave, **digital systems**
UHF (300–3000 MHz)	1 m–10 cm	Radar, TV, cell phones, **digital systems**
VHF (30–300 MHz)	10 m–1 m	TV, FM radio, police radio, **digital systems**
HF (3–30 MHz)	100 m–10 m	Shortwave radio (ham), citizens band
MF (300–3000 kHz)	1 km–100 m	AM broadcasting, maritime radio
LF (30–300 kHz)	10 km–1 km	Loran navigation, ADF beacons
VLF (3–30 kHz)	100 km–10 km	Long-range navigation, sonar
ELF (3–30 Hz)	62,137 mi–6214 mi	Detection of buried metal objects

Knowing the spectrum distribution of electronic systems helps identify interference sources.

Relative Permittivities of Various Dielectrics ***(chap1.pdf, p.23)***

Material	ϵ_r
Air	1.0005

Material	ϵ_r
Styrofoam	1.03
Teflon	2.1
Polyethylene	2.3
Polystyrene	2.5
Nylon	3.5
PVC	3.5
Epoxy resin	3.6
Quartz (fused)	3.8
Mylar	4.0
Epoxy glass (PCB substrate)	**4.7**
Bakelite	4.9
Glass (pyrex)	5.0
Silicon	12.0

Relative Permeabilities and Conductivities of Metals *(chap1.pdf, p.24)*

(Relative to copper)

Conductor	σ_r	μ_r
Silver	1.05	1
Copper-annealed	1.00	1
Gold	0.70	1
Aluminum	0.61	1
Nickel	0.20	600
Steel (SAE 1045)	0.10	1000
Stainless Steel (430)	0.02	500
Mumetal (at 1 kHz)	0.03	30,000

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6. Decibels and Common EMC Units

Power Gain *(chap1.pdf, p.25)*

$$\text{Power gain} \equiv \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{v_{\text{out}}^2/R_L}{v_{\text{in}}^2/R_{\text{in}}}$$

(chap1.pdf, p.25)

where $P_{\text{in}} = v_{\text{in}}^2/R_{\text{in}}$ and $P_{\text{out}} = v_{\text{out}}^2/R_L$.

Power gain in decibels:

$$\text{Power gain}_{\text{dB}} \equiv 10\log_{10}\left(\frac{P_{\text{out}}}{P_{\text{in}}}\right)$$

(chap1.pdf, p.25)

Figure description: Amplifier circuit with source V_S, source resistance R_S, input resistance R_{in}, amplifier block, load R_L.

Voltage and Current Gains *(chap1.pdf, p.26)*

$$\text{Voltage gain} \equiv \frac{v_{\text{out}}}{v_{\text{in}}}, \qquad \text{Current gain} \equiv \frac{i_{\text{out}}}{i_{\text{in}}}$$

In dB:

$$\text{Voltage gain}_{\text{dB}} \equiv 20 \log_{10}\left(\frac{v_{\text{out}}}{v_{\text{in}}}\right)$$

$$\text{Current gain}_{\text{dB}} \equiv 20 \log_{10}\left(\frac{i_{\text{out}}}{i_{\text{in}}}\right)$$

(chap1.pdf, p.26)

When $R_{\text{in}} = R_L$:

$$\text{Power gain} = \left(\frac{v_{\text{out}}}{v_{\text{in}}}\right)^2_{R_L=R_{\text{in}}}, \qquad \text{Power gain}_{\text{dB}} = 20 \log_{10}\left(\frac{v_{\text{out}}}{v_{\text{in}}}\right)_{R_L=R_{\text{in}}}$$

(chap1.pdf, p.26)

This equals the voltage gain (and similarly the current gain via $P = i^2 R$).

Conversion Table *(chap1.pdf, p.27)*

Ratio	V or I in dB	P in dB
10^6	120	60
10^4	80	40
10^2	40	20
10	20	10
2	6.02	3.01
1	0	0
10^{-1}	-20	-10
10^{-2}	-40	-20

Using dB in Amplifier Chains ***(chap1.pdf, p.28)***

$$P_{\text{out,dB}} = \text{Gain}_{\text{dB}} + P_{\text{in,dB}}$$

$$P_{\text{out,dBm}} = \text{Gain}_{\text{dB}} + P_{\text{in,dBm}}$$

$$P_{\text{out,dB}\mu\text{W}} = \text{Gain}_{\text{dB}} + P_{\text{in,dB}\mu\text{W}}$$

When $R_{\text{in}} = R_L$:

$$v_{\text{out,dB}} = \text{Gain}_{\text{dB}} + v_{\text{in,dB}}$$

$$i_{\text{out,dB}} = \text{Gain}_{\text{dB}} + i_{\text{in,dB}}$$

(chap1.pdf, p.28)

> Gain and Loss are opposite in sign. (dBmV, dBμV) and (dBmA, dBμA) also apply.

Power Loss in Cables — Lossy Transmission Line ***(chap1.pdf, p.29)***

For a lossy transmission line (matched, $Z_L = Z_C$):

$$\hat{V}(z) = \hat{V}^{+} e^{-\alpha z} e^{-j\beta z} = \hat{V}_f(z)$$

$$\hat{I}(z) = \frac{\hat{V}^{+}}{Z_C} e^{-\alpha z} e^{-j\beta z} = \frac{\hat{V}_f(z)}{Z_C}$$

(chap1.pdf, p.29)

Average power delivered:

$$P_{\text{av}}(z) = \frac{1}{2}\,\mathcal{R}]\,[\hat{V}(z)\hat{I}^{*}(z)] = \frac{1}{2}\frac{V^{+2}}{Z_C} e^{-2\alpha z} \cos\theta_{Z_C}$$

(chap1.pdf, p.29)

Power Loss Definition ***(chap1.pdf, p.30)***

$$\text{Power loss} = \frac{P_{\text{av}}(z=0)}{P_{\text{av}}(z=\mathscr{L})} = \frac{P_{\text{in}}}{P_{\text{out}}} = e^{2\alpha\mathscr{L}}$$

(chap1.pdf, p.30)

Cable loss in dB:

$$\text{Cable loss}_{\text{dB}} = 10\log_{10}(e^{2\alpha\mathscr{L}}) = 20\alpha\mathscr{L}\log_{10} e = 8.686\,\alpha\mathscr{L}$$

(chap1.pdf, p.30)

Therefore the **attenuation constant** can be found from measured loss:

$$\alpha = \frac{\text{power loss in dB/length}}{8.686\,L}$$

(chap1.pdf, p.30)

Important relationship when $R_L = 50\,\Omega$:

$$\text{dB}\mu\text{V (RMS)} = 107 + \text{dBm}$$

(chap1.pdf, p.30)

Signal Source and Measurement Instruments ***(chap1.pdf, p.31)***

- Signal sources modeled as **Thevenin equivalent** with $R_S = 50\,\Omega$ (virtually all modern sources).
- **Spectrum analyzer**: $C_{\text{in}} = 0$, $R_{\text{in}} = 50\,\Omega$
- **Oscilloscope**: $C_{\text{in}} = 47\,\text{pF}$, $R_{\text{in}} = 1\,\text{M}\Omega$

Matched Measurement System ***(chap1.pdf, p.32–33)***

Figure description: Signal source (50 Ω) connected via coaxial cable ($\hat{Z}_C = 50\,\Omega$) to signal measurer (50 Ω).

When the load equals Z_C (matched), the input impedance **at any frequency and for any cable length** is $Z_{\text{in}} = 50\,\Omega = Z_C$. This is why modern EMC test equipment uses pure 50 Ω impedances and 50-Ω coaxial cables — it enables **swept-frequency measurements** with constant output.